Flow Matching Policy Optimization with Mirror Descent and Entropy Constraints

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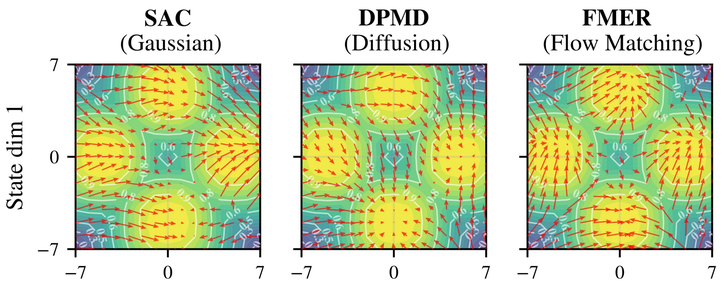
TL;DR FMER uses **Flow Matching (ODE)** as the policy generator, turns policy mirror descent into a **weighted flow-matching regression** and adds an **exact, tanh-corrected entropy constraint**, giving the best results on sparse-reward FrankaKitchen.

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What is the problem?

Good behaviour is often multimodal

Gaussian policies pick one of several equally good actions.



2D multi-goal task (bright = goals, arrows = chosen actions). SAC and DPMD drift to favourite goals; FMER heads for the nearest of four equally good goals.

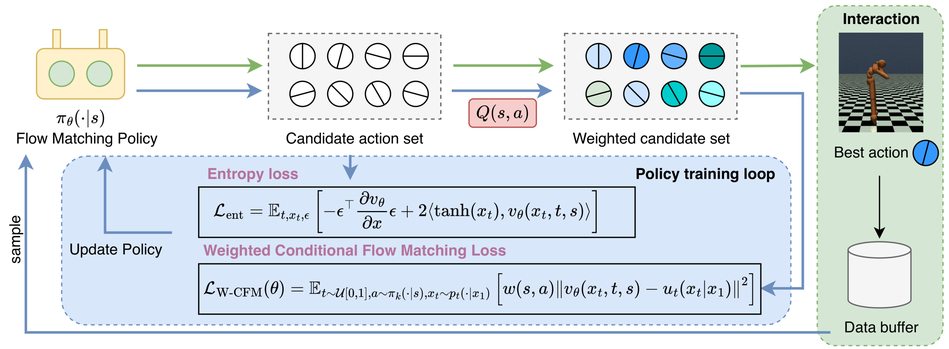
What we want

- ❀ **Joint optimization: policy mirror descent + entropy constraint**
- ❀ **Multimodal expressivity**
- ❀ **No backprop through generation chain**
- ❀ **Non-heuristics entropy estimation**

→ Flow Matching Policy

	Mirror descent	Entropy constraint	Unbiased entropy	Simulation-free
DIPO	✗	✗	✗	✓
DPMD	✓	✗	✗	✓
QVPO	✗	✓	✗	✓
SAC-Flow	✗	✓	✗	✗
FLAME	✗	✓	✗	✗
FMER (ours)	✓	✓	✓	✓

FMER: one simulation-free update



A **policy mirror descent** step and a **minimum-entropy constraint**:

$$\mathcal{L}_{\text{FMER}}(\theta) = \underbrace{\mathcal{L}_{\text{W-CFM}}(\theta)}_{\text{Exploitation}} + \underbrace{\alpha\mathcal{L}_{\text{ent}}(\theta)}_{\text{Exploration}}$$

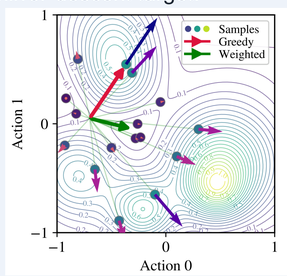
Mirror descent = weighted flow matching

Theorem 1. Minimisers of the weighted CFM loss = mirror-descent target:

$$\mathcal{L}_{\text{W-CFM}} = \mathbb{E}_{t,s,a,x_0}\left[w(s,a)\,\|v_\theta(x_t,t,s) - (x_1 - x_0)\|^2\right]$$

$$\Longrightarrow \quad \pi_{\theta^*}(a|s) = \frac{\pi_k(a|s)\exp(A(s,a)/\tau)}{Z(s)}$$

- Sample M candidates to approximate $MZ(s)$
- Pure regression: **no backprop through the ODE**
- Soft weights pool all good candidates



Analytical entropy in a bounded action space

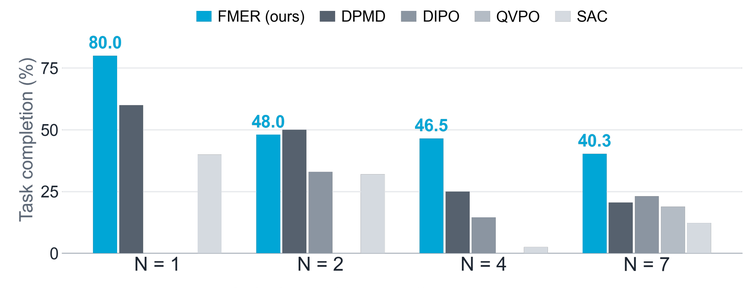
Entropy = integrated divergence of the velocity field; for bounded actions a = tanh(x), **Prop. 1** adds the exact **tanh correction**:

$$\mathcal{L}_{\text{ent}} = -\,\mathbb{E}_{t,x_t}\left[\nabla_x \cdot v_\theta \;-\; 2\left\langle \tanh(x_t),\, v_\theta \right\rangle\right]$$

Does it work?

Sparse-reward FrankaKitchen

Reward only on sub-task completion · 280 steps · 1M steps · 5 seeds



Not shown: FLAME, TD3, PPO, SAC-Flow (all ≤ 18.3%).

1.7× best baseline, FrankaKitchen N = 7, 40.3% vs 23.1% (DIPO)

1.9× best baseline, FrankaKitchen N = 4, 46.5% vs 25.0% (DPMD)

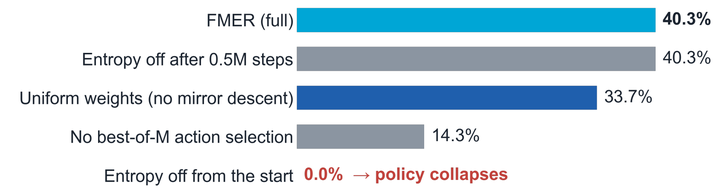
FMER completes **5 tasks** in 280 steps (best baseline DIPO: 3)



Dense-reward MuJoCo

HalfCheetah	Walker2d	Humanoid	Hopper
12.3k	**5.28k**	**5.29k**	**2.91k**
#1 of 9	#1 of 9	#2 of 9	#7 of 9

Ablation studies (FrankaKitchen N = 7)



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